

- 3 a** Simple harmonic motion (SHM) is defined as the periodic motion of an object in which its acceleration is directly proportional to its displacement from a fixed point and acting in the opposite direction to its displacement. **B1**
- bi** $f = \frac{1}{T} = \frac{1}{0.50} = 2.0 \text{ Hz}$ **A1**
- bii** $a = -\omega^2 x \rightarrow \omega^2 = \left(\frac{\rho Ag}{m} \right)$ **C1**
- $$\rightarrow 4\pi^2 f^2 = \left(\frac{\rho Ag}{m} \right) \rightarrow \rho = \frac{4\pi^2 f^2 m}{Ag}$$
- $$\rho = \frac{4\pi^2 (2.0)^2 0.032}{4.2 \times 10^{-4} (9.81)}$$
- C1**
- $$\rho = 1230 \text{ kg m}^{-3}$$
- A1**
- biii** $E = \frac{1}{2} m \omega^2 x^2$
- $$E_{t=0} = \frac{1}{2} m \omega^2 x_0^2 = \frac{1}{2} 0.032 (4\pi^2 2.0^2) (-1.5 \times 10^{-2})^2 = 5.68 \times 10^{-4} \text{ J}$$
- C1**
- $$E_{t=1.50 \text{ s}} = \frac{1}{2} m \omega^2 x_{1.50}^2 = \frac{1}{2} 0.032 (4\pi^2 2.0^2) (-0.65 \times 10^{-2})^2 = 1.07 \times 10^{-4} \text{ J}$$
- C1**
- $$\% \text{ loss} = \frac{5.68 \times 10^{-4} \text{ J} - 1.07 \times 10^{-4} \text{ J}}{5.68 \times 10^{-4} \text{ J}} = 81\% \text{ loss}$$
- A1**
- biv** As the liquid is more viscous, greater damping forces will act on the tube. **M1**
- More negative work done on the tube per oscillation (rate of energy loss greater). **M1**
- Hence, the tube oscillates with much less energy in this liquid.
- Energy is directly proportional to the square of the amplitude. Therefore, the tube oscillates at smaller amplitudes. **A0**